<u>**PROJECT 08 REPORT**</u>

Servo Motor Position Control System using SG90 Servo Motor

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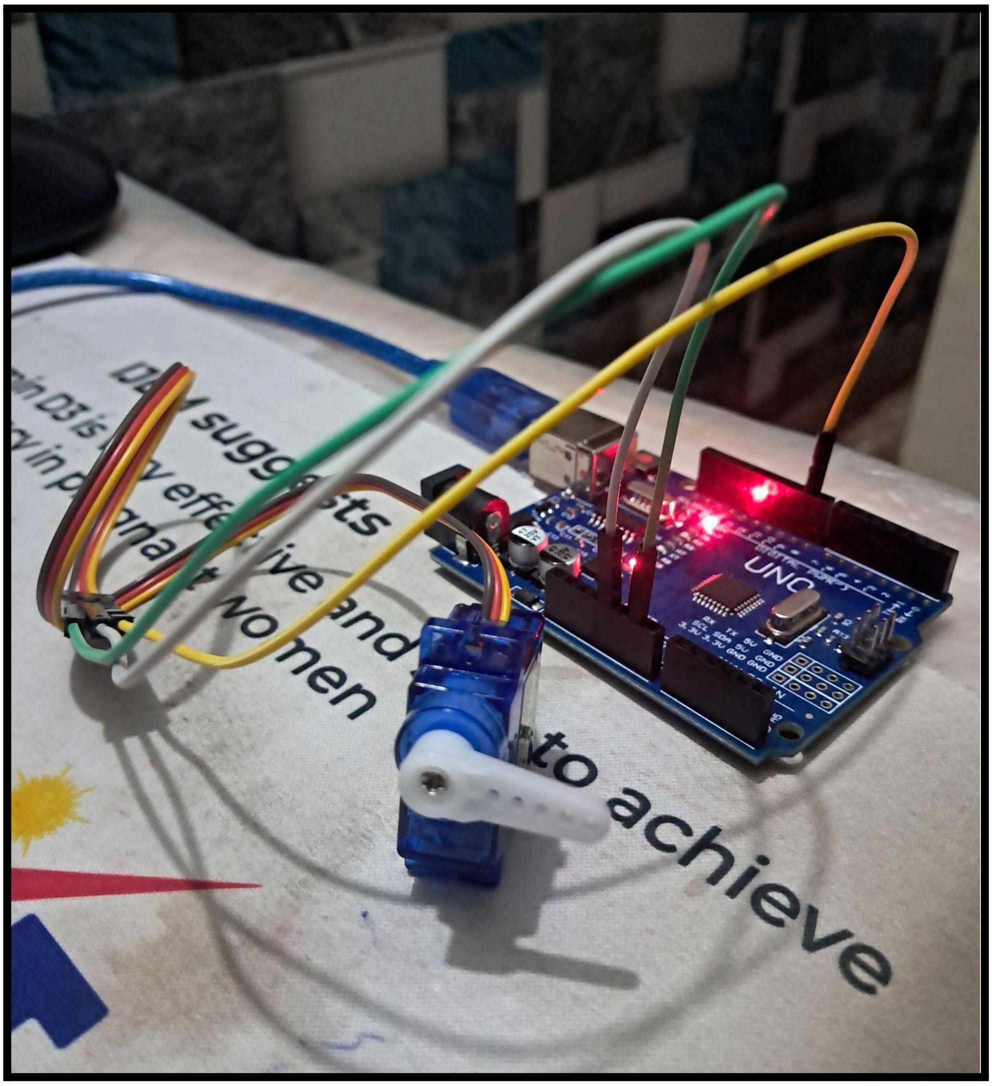
Duration : 2 days



1. Objective

The objective of this project was to understand the working principle of the SG90 micro servo motor and interface it with Arduino Uno for accurate angular position control.

The project demonstrates how a servo motor can rotate to specific angles as commanded by Arduino

2. Components Used

Component	Quantity
Arduino Uno R3	1
SG90 Micro Servo Motor	1
Jumper Wires	Multiple
USB Cable	1

3. Software Used

- Arduino IDE
- Servo Library (Servo.h)
- Windows Laptop

4. Theory

The SG90 is a 9 gram micro servo motor capable of rotating approximately 0° to 180° with high positional accuracy.

Unlike a normal DC motor that rotates continuously, a servo motor rotates to a commanded position and holds that position using an internal feedback system.

The SG90 consists of:

- DC Motor
- Gear Train
- Control Circuit
- Potentiometer (Position Feedback)

The servo has three connecting wires:

- Brown Wire → Ground (GND)
- Red Wire → +5V Supply
- Orange/Yellow Wire → PWM Control Signal

The internal control circuit continuously compares the commanded position with the current position obtained from the potentiometer and rotates the motor until both positions match.

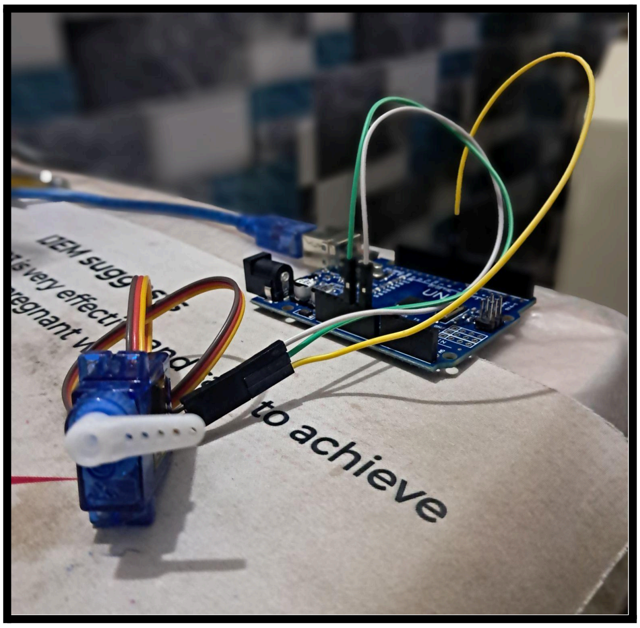
This continuous comparison between the desired angle and the current angle forms a closed loop feedback system.

5. Tasks Performed

➢ Understanding the SG90 Servo Motor

The SG90 micro servo motor was studied to understand its construction, operating principle, 3 wire interface, internal gear mechanism, servo horns, mounting hardware and closed loop feedback system.

➢ Basic Servo Position Control



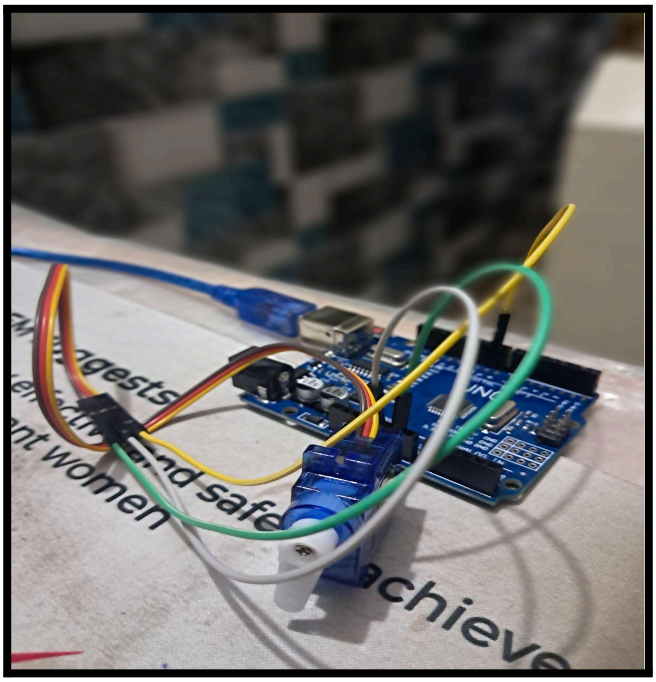
Video

The SG90 servo motor was interfaced with Arduino Uno using the Servo library.

The servo was commanded to rotate to fixed angular positions of: **0° → 90° → 180°**

The servo successfully rotated to each commanded angle and maintained its position until the next command was received.

➢ Smooth Servo Sweep



Video

The servo movement was improved using for loops to generate smooth angular transitions instead of abrupt position changes.

Servo operation:

0° → 1° → 2° → ... → 180° → 179° → 178° → … → 0°

Small delays were introduced between successive angle commands to produce smooth mechanical motion.

6. Learning Outcomes

During the making of this project, the following concepts were learned:

- SG90 Servo Motor Construction
- Servo Motor Operation
- Difference Between Servo Motor and DC Motor
- Actuators in Embedded Systems
- PWM Based Position Control
- Servo Library (Servo.h)
- Servo Object Creation
- attach() Function
- write() Function
- Closed Loop Control
- Internal Potentiometer Feedback
- Angular Position Control
- Smooth Motion using for Loops
- Mechanical Assembly and Servo Horn Installation

7. Problems Faced

I. The servo horn occasionally appeared to skip intermediate positions or move abruptly.**(The horn was properly fixed using the supplied center screw)**

II. Recording the demonstration video was difficult because the lightweight servo could not remain upright on its own.**(The servo was manually stabilized during recording to clearly demonstrate the rotational movement)**

8. Observations

- The SG90 servo accurately rotated to commanded angular positions.
- The servo maintained its position after reaching the desired angle.
- Smooth motion was achieved by gradually changing the angle using for loops.
- Larger delay values produced slower and smoother motion.
- The servo generated a small buzzing sound while positioning and holding its angle.
- Proper horn installation was necessary for accurate movement observation.
- The servo library greatly simplified PWM based servo control.

9. Applications and Future Scope

Servo motors are widely used wherever accurate angular positioning is required such as in Robotic Arms,Automatic Door Systems and Smart Dustbins.

Possible future upgrade includes Ultrasonic Radar Scanner,Multi Servo Robotic Arm and Ultrasonic SensorBased Smart Dustbin.

10. Conclusion

This project successfully demonstrated precise angular position control using the SG90 micro servo motor and Arduino Uno.

The project represents the first **actuator control system** in the portfolio and provides a strong foundation for future robotics, automation, smart mechanisms and sensor actuator integrated embedded systems.